SHROOMING WITH STATISTICAL LEARNING

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Abstract. Statistical learning methods are applied to a dataset with categorical predictors and a binary response. The data set under study is the Mushroom Classification dataset provided by UCI Machine Learning at this link.

1. Introduction

A mushroom is the part of a fungus that contains reproductive units or "spores." Fungi differ from plants in their methods of acquiring food. Fungi rely on the organic material immediately surrounding them to obtain nourishment, while plants are capable of converting sunlight and carbon dioxide to food through the process of photosynthesis. Similar to plants, mushrooms come in many types and are sometimes edible. In some cases, consumption of a mushroom can be fatal. Contrary to the canonical example of poison ivy being identifiable with the adage "leaflet of three, let it be," there is no straightforward method for identifying a random mushroom as poisonous or not. For this reason, machine learning methods may be useful for classifying the edibility of a mushroom when several characteristics are considered.

The data set under study originally comprised a hypothetical sample of mushrooms classified in [1] as either edible, poisonous or suspect. All "suspect" observations have been converted to "poisonous" here. There are $p = 22$ categorical predictor variables of whether or not a given mushroom is poisonous or edible, with $n = 8124$ observations. The observations have been split half-and-half into training and testing sets. All misclassification rates reported are for the test sets. One of the predictor variables, "veil-type", contained the same value for all observations and thus provided no information. For this reason, "veil-type" was omitted from the statistical analysis making $p = 21$. The two genera of mushroom contained in this data set are Agaricus and Lepiota. These two genera both belong to the same family Agaricaceae.

2. Goals, Motivations, Challenges

The goal of this research is to find patterns that would help to guide one toward general principles of accurate identification of a mushroom as either poisonous or edible. Besides the possibility of death from eating the wrong mushroom, there also exist benefits to eating the right mushroom. Cultures around the globe have been using mushrooms as medicine and food for thousands of years [2]. Contemporary research shows that this is for good reason; many mushrooms possess numerous potent health benefits. In particular, studies have shown that consumption of medicinal mushrooms provides anti-inflammatory, anti-viral, anti-oxidant, anti-cancer, and anti-microbial benefits that contribute toward preventing a plethora of diseases [3]. Indeed, an entire scientific journal titled "International Journal of Medicinal Mushrooms" was created in 1999 to conduct research in this area.

Date: March 24, 2026.

One challenge that this data set poses is that none of the variables are numeric. All predictors are categorical here. A few of the variables are ordinal, like "gill-size", "gill-spacing", "ring-number", and "population." The other 17 predictors are all nominal. In section 3, techniques are discussed for adapting this data set for use with the non-parametric K-nearest neighbors method.

3. Methods

Decision trees, logistic regression, random forests, support vector machines, neural networks, linear discriminant analysis, K-nearest neighbors, bagging, and boosting are all suitable methods for a classification problem such as this. Decision trees have the advantage of being simple and easy to interpret, but possibly possess high variance. When the number of possible values of an unordered predictor q is large, computational costs may become high for decision trees. Since the largest q is 7 for this data set, this is not an issue here. Logistic regression and linear discriminant analysis are algorithmically simple for the case of binary responses, but assume a linear decision boundary. Bagging improves upon the high variance of decision trees, but is harder to interpret and correlates the trees. Random forests decorrelates the trees by use of a random subset of candidate variables at each split in the tree, but are again tougher to interpret than a single decision tree. Boosting decision trees are advantageous if overfitting is an issue, but suffer the potential disadvantage of being difficult to properly tune. Support vector machines overcome the limitations of the linear decision boundary assumption of LDA and logistic regression, as well as the overfitting issue inherent to boosting decision trees. Single hidden layer back-propagation neural networks, which are the ones reported on in subsection 4.4, enjoy local information transmission to and form adjacent nodes but can be complex to train well. K-nearest neighbors is flexible, but often suffers high variance if K is too small [4, 5].

All of the methods mentioned above are used here. Decision trees are used on odored and odorless mushrooms separately. These trees provide a sequence of properties to inspect if encountering mushrooms of Agaricus or Lepiota without a computer and machine learning capabilities. The other methods used serve as a proof of concept that statistical learning can be used quite effectively for classifying mushrooms of these genera as either edible or poisonous. From a purely mathematical perspective, the results below are a demonstration that learning methods can perform well for categorical predictors of a binary response.

When applying K-nearest neighbors to categorical predictors, one must define a distance metric other than Euclid's or code the categorical predictors using dummy variables. The latter technique is used here. For example, the predictor variable "cap-shape" has 6 possible character-type values:

(1) 'b'
(2) 'c'
(3) 'x'
(4) 'f'
(5) 'k'
(6) 's'

To code "cap-shape" using dummy variables, I have created 6 new variables with the names

(1) cap-shapeb
(2) cap-shapec
(3) cap-shapef
(4) cap-shapek
(5) cap-shapes
(6) cap-shapex

These 6 new variables together replace "cap-shape" and follow the coding scheme that, for observation i:

$$\text{cap-shapeb}_i = \begin{cases} 1, & \text{if cap-shape}_i = \text{b} \\ 0, & \text{if cap-shape}_i \neq \text{b} \end{cases}$$

$$\text{cap-shapec}_i = \begin{cases} 1, & \text{if cap-shape}_i = \text{c} \\ 0, & \text{if cap-shape}_i \neq \text{c} \end{cases}$$

$$\text{cap-shapex}_i = \begin{cases} 1, & \text{if cap-shape}_i = \text{x} \\ 0, & \text{if cap-shape}_i \neq \text{x} \end{cases}$$

$$\text{cap-shapef}_i = \begin{cases} 1, & \text{if cap-shape}_i = \text{f} \\ 0, & \text{if cap-shape}_i \neq \text{f} \end{cases}$$

$$\text{cap-shapek}_i = \begin{cases} 1, & \text{if cap-shape}_i = \text{k} \\ 0, & \text{if cap-shape}_i \neq \text{k} \end{cases}$$

$$\text{cap-shapes}_i = \begin{cases} 1, & \text{if cap-shape}_i = \text{s} \\ 0, & \text{if cap-shape}_i \neq \text{s} \end{cases}$$

This dummy variable coding has been done for all of the predictor variables and used for methods that require quantitative predictors.

4. Results

4.1. **Random forests.**

The results for random forests where the number of candidate variables considered at each split is $m = \sqrt{p} = \sqrt{21} \approx 5$ are shown in table 1. For any number of trees used that is greater than 3, a misclassification rate of 0 results.

Number of trees	1	2	> 3
Misclassification rate	0.001	0.0007	0

Table 1. Random forests results.

Variable importance, as measured by the mean decrease in Gini index, is shown in figure 1. It is evident that "odor" is the most important variable. The possible values of "odor" are:

(1) Almond
(2) Anise
(3) Creosote
(4) Fishy
(5) Foul
(6) Musty
(7) None

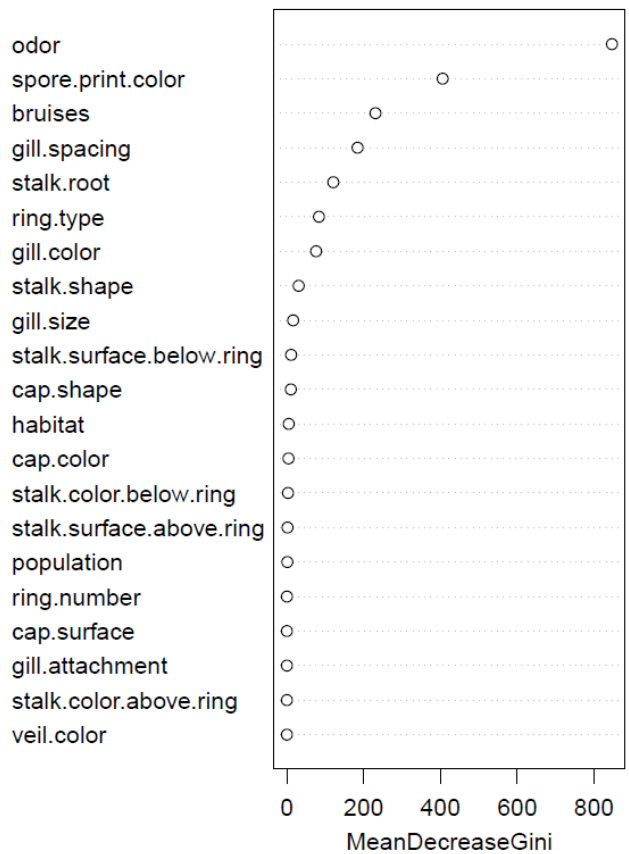

FIGURE 1. Variable importance plot for random forests

The percentage of observations with odor=None is 43%, while the percentage of observations with odor≠None is 57%. The data has been partitioned into odorless and odored observations for the following subsection 4.2.

4.2. **Decision trees.**

The decision tree obtained on the odored observations is shown in figure 2. "Anise" is a word used to describe the smell of fennel or licorice. "Creosote" is a word used to describe the smell of tar. It appears that smells generally associated with things humans eat emanate from edible mushrooms. This seems a rather subjective measure. Perhaps a more accurate diagnostic would involve performing some chemical analysis.

When running a decision tree on the odorless observations, the structure shown in figure 3 is acquired. It appears that shrooms with green or purple spores, regardless of bruise presence, stalks, or gills, are almost certainly poisonous. On the other hand, shrooms with other spore colors need to be inspected with more care before concluding their edibility.

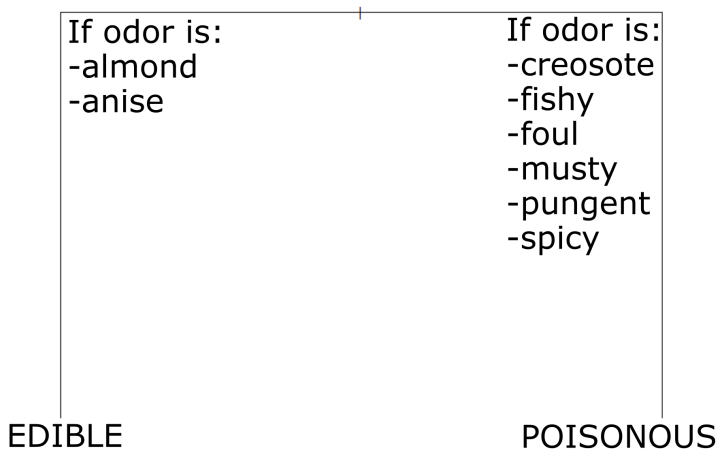

FIGURE 2. Decision tree for odored mushrooms

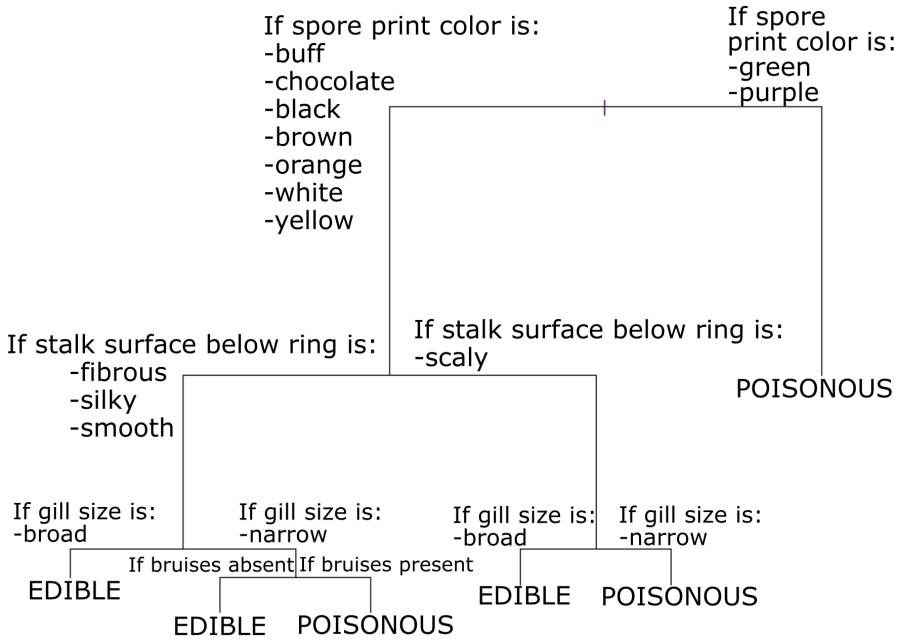

FIGURE 3. Decision tree for odorless mushrooms

If the stalk surface below the ring is scaly, all one needs to check further is gill size. If the stalk surface below the ring is fibrous, silky, or smooth, one must check the gill size and bruise presence before concluding the edibility.

4.3. **K-nearest neighbors (KNN).**

As mentioned in section 4, the dummy variable encoding was used for KNN. The results are shown in figure 4. Misclassification rate is lowest when K is low. This is reflective of the fact that the indicator value used for the encoding of the categorical predictors is 1 and not some other larger value. In other words, the dummy variable encoding used results in the predictors being rather close to one another in p-dimensional space, when closeness is measured by Euclidean distance. Thus, a small K gives the best results.

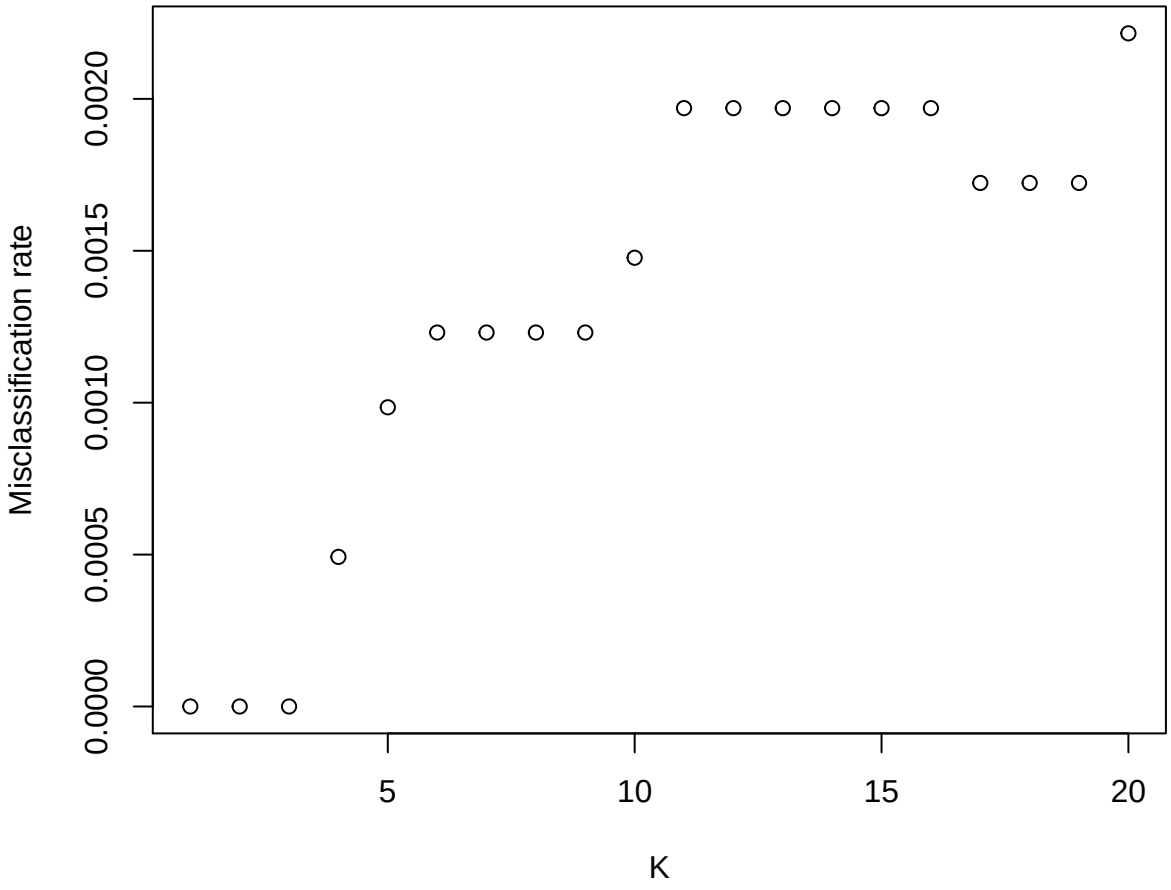

FIGURE 4. Misclassification rate as a function of K, the number of nearest neighbors considered.

4.4. **Other methods: all variables considered.**

The results of all other methods used when all variables are included are summarized in table 2.

Method	Misclassification rate
Random Forests w/ ntrees=3	0
SVM w/ linear kernel	0
0 hidden layers N-net, decay=0	0
1 hidden layer w/ 1 node N-net, decay=0	0.0118
1 hidden layer w/ 1 node N-net, decay=0.05	0
1 hidden layer w/ 5 nodes N-net, decay=0.5	0
Boosting w/ ntrees=100, depth=4	0
Bagging	0
Linear Discriminant Analysis (LDA)	0.0002
Logistic Regression	0

Table 2. Results of other methods. "ntrees" stands for number of trees. "SVM" stands for support vector machine. "N-net" stands for neural network.

The methods in table 2 that yield a non-zero misclassification rate have been investigated further. The confusion matrices for these methods are shown in tables 3 and 4. The neural net reported on in table 3 would have deadlier consequences if used on actual mushroom data than would the LDA reported on in table 4. The reason for this is that 48 poisonous mushrooms have been classified as edible in the case of the neural net. On the other hand, LDA predicts that 1 of the edible mushrooms is poisonous; not nearly as fatal an error.

		True edibility	
		e	p
Predicted edibility	e	2130	48
	p	0	1884

Table 3. Confusion matrix for single-layer, single-node neural net with decay=0. 'e' stands for edible, 'p' stands for poisonous.

		True edibility	
		e	p
Predicted edibility	e	2130	0
	p	1	1931

Table 4. Confusion matrix for linear discriminant analysis.

4.5. **Logistic regression with subsets of important variables.**

Some of the perfect results reported in table 2 of subsection 4.4 may be the result of overfitting. After coding all predictors using dummy variables, there are 105 predictor variables. While this is still significantly smaller than the number of total observations, overfitting becomes more of an issue for some of the methods used here as p gets closer to n. In this subsection, logistic regression is used on subsets of the predictor variables, guided by the importance plot in figure 1 and the decision trees of figures 2 and 3.

Table 5 shows the confusion matrix of a logistic regression where the spore print color of odorless mushrooms is the only variable considered. The misclassification rate is 0.48 (specificity and sensitivity here too).

		True edibility	
		e	p
Predicted edibility	e	2101	1921
	p	29	11

Table 5. Confusion matrix for logistic regression on odorless mushrooms where spore print color is the only predictor.

Table 6 shows the confusion matrix of a logistic regression where the odor of odored mushrooms is the only variable considered. The misclassification rate is 0.43. The fact that this is lower than the misclassification rate of the results in table 5 is consistent with the difference in decision tree complexity between figures 2 and 3.

		True edibility	
		e	p
Predicted edibility	e	382	0
	p	1748	1932

Table 6. Confusion matrix for logistic regression on odored mushrooms where odor is the only predictor.

Based on the asymmetries in tables 5 and 6, it appears that the presence of odor is a good indicator of whether or not the majority of Agaricus and Lepiota mushrooms are poisonous. However, of the odorless mushrooms, it is important to check more than just the spore print color to determine whether or not a mushroom is poisonous. If one were to follow the predictions of the model reported on in table 5, there is a about a 50/50 chance of eating the wrong mushroom. The logistic regression reported on in table 6 would be much better to follow, since it is very conservative. None of the poisonous mushrooms are classified as edible, but a majority of the edible mushrooms are classified as poisonous. This indicates that, if odor is the only variable considered, one will correctly classify many mushrooms as poisonous but is less likely to correctly classify a mushroom that can be eaten safely.

4.6. **Neural Networks with subsets of important variables.** A variety of neural networks have been performed on the full data set, as well as subsets of the full data set. Table 7 shows the results for a neural network with two hidden layers. This network has been provided two predictor variables of the dummy data set: "odorn" and "spore.print.colorw". More clearly, this means that the model is informed on whether a mushroom has odor as well as if its spore print color is white or not. It does quite well for having so little information, achieving a misclassification rate of 0.11.

		True edibility	
		e	p
Predicted edibility	e	1748	70
	p	382	1862

Table 7. Confusion matrix for neural network with two hidden layers, informed on the presence of odor and white spores.

Table 8 reports on the performance of a neural network with two hidden layers, informed on 4 predictor variables in the dummy variable data set: "odora," "odorl," "odorc," and "odory." In other words, this model knows whether a given mushroom has an anise, almond, creosote, or fishy smell. The performance of this network is worse than the one from table 7 (misclassification rate from results of table 8 is 0.389), indicating that the presence of an odor is more predictive of toxicity than the odor itself.

		True edibility	
		e	p
Predicted edibility	e	2130	1581
	p	0	351

Table 8. Confusion matrix for neural network with two hidden layers, informed on 4 different possible odors that an odored mushroom can have.

Interestingly, the threshold used for the partial derivatives of the error function and the number of repetitions for neural network training has little effect on the two results above. Additionally, the activation function doesn't appear to matter much either. When using the softplus activation function instead of the default logistic activation function for the model table 7 derives from, a misclassification rate of 0.11 still results.

5. Closing Remarks and Future Outlook

Statistical learning has proved effective at classifying gilled mushrooms in the Agaricus and Lepiota genera when many characteristics are considered. The decision tree method used in subsection 4.2 is instructive in regards to what factors one should consider for odored mushrooms relative to odorless mushrooms. The variable importance plot from the random forests method displayed in figure 1 shows the hierarchy of characteristics to consider. Interestingly, the presence of bruises is relevant for predicting edibility but cap-color is relatively unimportant.

The results of subsections 4.5 and 4.6 show that the presence of odor alone is more important than the specific odor when predicting edibility. Among the odorless mushrooms, consideration of spore print color is necessary for identifying many of the edible mushrooms; but one must consider other variables if also looking to avoid the poisonous odorless mushrooms.

Neural networks with two hidden layers appear to be a quality method to use if only a few variables are known, like the presence of odor and spore print color. There is room for improvement from the results of table 7, likely in the realm of tuning the number of nodes in the hidden layers.

All of the mushrooms in the data set under consideration here are gilled, meaning that they have fan-like structures underneath their cap. In practice, the spore print color is obtained by pressing the cap, gilled-face down onto some suitable surface, covering it with a cup and leaving it overnight. The color of the print one discovers in the morning is the "spore-print-color" as reported in this data set. This variable is more important to consider than the gill size, spacing, color, or attachment. It appears that the spores of the organism contain the compounds that may or may not be poisonous. Perhaps it is these same spores that give off a distinct odor for some mushrooms, but not for others. The families of mushrooms considered here belong to the larger class *Basidiomycetes*. Different classes of mushroom differ in the way they reproduce, and have different structures that are associated with reproduction [1]. It seems reasonable to think that the life cycle of the organism determines its spore content. This may indicate that some mushrooms are edible during some point in time but become poisonous at different times.

References

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